

Fontys University of Applied Sciences

Resume Analyzer

Data Challenge – AI Advanced

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Idea

The Idea

The idea is to create an AI-powered career assistant that analyzes resumes and helps users improve them based on industry standards. The assistant will detect missing sections, highlight weak areas, and provide personalized suggestions using deep learning techniques such as Named Entity Recognition (NER) and language modeling.

Problem and Relevance

Many students and job seekers find it difficult to craft resumes that stand out to recruiters and applicant tracking systems (ATS). Feedback is often hard to come by, and many people don't realize how their resumes are being automatically screened or rejected by AI systems used in hiring processes. This tool will help users tailor their resumes effectively and feel more confident in their job search.

Data

To support the full vision of the Resume Analyzer as a smart career assistant, several types of data would be valuable. First, a large and diverse corpus of resumes from various fields, industries, and experience levels is essential to understand the structure, content, and language commonly used in successful applications. Ideally, this data would also include information about which resumes led to interviews or hires, so the model can learn what makes a resume effective—not just well-structured. While some of this data is not publicly available, it defines the ideal dataset for achieving the broader goal of personalized, data-driven career coaching.

Innovation and Impact

This project is innovative because it blends deep learning with career development, offering users:

- A resume “health check” before applying to jobs
- Instant, structured feedback similar to what companies use internally

Additionally, many companies now use AI to screen resumes, filtering them before a human ever sees them. This tool provides users with a comparable AI-based perspective, helping them improve their chances of making it past those filters.

Ethical Considerations

Key ethical aspects include:

- **Data Privacy:** Resume content may contain sensitive personal information. Any data submitted through the MVP will not be stored or shared.
- **Bias:** AI models trained on biased data could favor certain education levels, job titles, or locations. I will monitor the model outputs and use fairness-aware scoring logic where possible.
- **Transparency:** Users will be informed of how the scoring and feedback are generated, avoiding any "black box" decisions.

Minimum Viable Product (MVP)

Functionality

The MVP will be a React-based web application that allows users to upload or paste their resume text. The backend will use a trained deep learning model to:

- Extract structured information (skills, education, etc.)
- Assign a resume quality score (1–10)
- Provide clear, categorized feedback on what’s good and what can be improved

The feedback categories may include:

- Skills coverage
- Experience and education depth
- Use of action verbs
- Resume length and formatting cues

Data Source

To build the MVP, I will use two complementary publicly available datasets:

1. Resume Entities for NER – DataTurks

This dataset contains resumes with annotated entities such as:

- Skills
- Education
- Degree
- Companies
- Designation

It is ideal for training a Named Entity Recognition (NER) model to extract structured information from resume text.

2. Resume Dataset – Gaurav Dutta

This dataset contains unlabeled resumes with category labels (e.g., Data Science, HR). It can be used for:

- Testing the entity extraction model on a variety of real-world examples
- Rule-based or scoring analysis
- Optional classification of resume domain

If needed, I will also generate synthetic resumes to increase diversity across industries, writing styles, and experience levels.

Tools and Techniques

- Deep Learning Models: BERT, RoBERTa (fine-tuned for NER and scoring logic)
- Libraries: Hugging Face Transformers, PyTorch or TensorFlow
- Frontend: React.js with Tailwind CSS
- Backend: FastAPI or Flask
- Data Preprocessing: Pandas, spaCy, NLTK
- Model Training: Jupyter Notebook

Definition of Done

- A user can submit a resume via the frontend
- The backend processes the resume, returns a score and meaningful feedback
- The analysis is based on extracted entities and follows basic evaluation rules
- The system reliably handles a variety of resume formats and content styles